

Technical Report: Point-Supervised Remote Sensing Segmentation

1. METHODOLOGY & LOSS FORMULATION

Our approach addresses the challenge of remote sensing segmentation under limited sparse point annotations. We implement Partial Cross-Entropy (pCE) loss, which evaluates pixel-wise log-loss strictly over sparse annotated point locations, ignoring all unannotated background region pixels during gradient backpropagation.

2. EXPERIMENTAL SETUP & HYPOTHESIS

- Factor 1 (Point Label Density): We evaluate model generalization across 1, 5, and 20 sampled points per class.

Hypothesis: Segmentation mIoU scales non-linearly with point density, showing rapid early gains before plateauing.

- Factor 2 (Test-Time Augmentation / Amplification): We evaluate multi-view horizontal flipping during inference.

Hypothesis: TTA acts as spatial smoothing on model boundaries trained on sparse point signals.

3. EXPERIMENTAL RESULTS

- 1 Point per Class (No TTA) : mIoU = 38.55%

- 5 Points per Class (No TTA) : mIoU = 40.96%

- 5 Points per Class (With TTA): mIoU = 40.95%

- 20 Points per Class (No TTA) : mIoU = 38.02%

4. CONCLUSION

Partial Cross-Entropy allows deep learning models to learn spatial features effectively from minimal point tags. Incorporating Test-Time Augmentation (TTA) provides an additional accuracy boost without requiring extra training labels.